

TECHNICAL DOCUMENTATION

Content:

- Title *Lightweight SNMP Agent for Autoca Embedded System*
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- Internship duration *6 months*
- University *OTH Regensburg*
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1. Abstract

This project focuses on the development of a lightweight SNMP (Simple Network Management Protocol) agent for the Autoca embedded controller used in industrial airport logistics systems. The primary objective was to enable remote monitoring and diagnostics of embedded devices using standard SNMP monitoring tools such as PRTG.

The implemented solution allows external systems to retrieve real-time system data using SNMP GET requests over UDP communication. A custom MIB structure was designed to store OID-based information, enabling efficient data access and response generation. The system ensures low resource utilization, making it suitable for embedded environments.

2. Introduction

This document describes the design, implementation, and validation of a lightweight SNMP (Simple Network Management Protocol) agent developed for the Autoca embedded system.

The purpose of this project is to enable remote monitoring of embedded devices in an industrial environment using standard SNMP-based monitoring tools.

The solution allows external systems (e.g., PRTG) to query device information such as system identity, runtime status, and heartbeat using SNMP GET requests.

3. Objective

The main objectives of the project were:

- Implement a basic SNMPv1 agent
- Enable UDP-based communication on port 161
- Design a custom MIB (Management Information Base)
- Support real-time monitoring of system parameters
- Ensure low resource usage suitable for embedded systems

4. System Architecture

4.1 Overview

The system follows a Manager-Agent architecture:

[SNMP Manager (PRTG / PC)]



[SNMP Agent (Autoca Controller)]



[MIB Database (OID → Value Mapping)]

4.2 Components

SNMP Manager

- External monitoring tool
- Sends SNMP requests

- Displays received data

SNMP Agent (Autoca)

- Embedded software module
- Responds to SNMP queries
- Retrieves data from MIB

MIB (Management Information Base)

- Internal database
- Stores:

OID → Value

5. Software Modules

The system is divided into modular components:

5.1 snmp_mib.c — Data Storage Layer

Purpose:

Stores all SNMP-readable data.

Structure:

- Fixed-size array (32 entries)
- Each entry:

OID (string)

Value (string)

Key Functions:

Initialization

SNMP_MIB_Init()

- Initializes MIB

- Enables system usage
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Add / Update

SNMP_MIB_AddOrUpdate()

- Adds new OID
 - Updates existing value
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Get Value

SNMP_MIB_Get()

- Searches OID
 - Returns corresponding value
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Safety Mechanism:

- **g_bReady** flag prevents:
 - access before init
 - access after shutdown
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5.2 snmp_udp.c — Network Communication Layer

Purpose:

Handles all network communication.

Responsibilities:

- Listens on UDP port 161
 - Receives SNMP requests
 - Sends responses
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Workflow:

1. Wait for incoming packet

2. Receive SNMP request
 3. Forward data to parser
 4. Send response back
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Reason for using UDP:

- Connectionless
 - Lightweight
 - Low overhead
 - Ideal for monitoring
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5.3 snmp_v1.c — Protocol Processing Layer

Purpose:

Handles SNMP protocol logic.

Responsibilities:

Decode incoming packet

- Extract OID from request

Encode response packet

- Build valid SNMP response

Handle errors

- Return noSuchName if OID not found
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Key Concept:

Uses BER (Basic Encoding Rules)

5.4 BER Encoding

Purpose:

Convert human-readable data to network format.

Rule:

[Tag][Length][Value]

Example:

Value:

"autoca 103"

Encoded:

04 0B 61 75 74 6F 63 61 20 31 30 33

Why needed:

- **Network uses binary data**
- **Ensures standardized communication**

5.5 snmp_tiny.c — Data Generation Layer

Purpose:

Provides dynamic system data.

Startup Initialization:

Adds fixed values:

- **System name**
 - **Location**
 - **Description**
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Runtime Behavior:

Every 2 seconds:

Update heartbeat counter

Example:

47 → 48 → 49 → ...

6. Multithreading Design

The system uses two threads:

Thread 1: Heartbeat

- Updates system status periodically

Thread 2: SNMP Listener

- Waits for incoming requests
- Handles communication

Benefit:

- Parallel execution
- Real-time response

7. Data Flow (End-to-End)

1. Manager sends OID request
2. UDP receives packet
3. SNMP parser extracts OID
4. MIB searches for value
5. Value returned to parser
6. Response encoded (BER)
7. UDP sends response

8. Example OIDs

OID	Description	Value
1.3.6.1.2.1.1.1.0	System Description	Beumer Autoca System
1.3.6.1.2.1.1.5.0	System Name	autoca 103
1.3.6.1.2.1.1.6.0	Location	Beckum
1.3.6.1.3.x.x	Heartbeat	Incrementing

9. Testing & Validation

Tools Used:

PRTG Network Monitor

Test Scenarios:

Test Case	Result
Valid OID request	Correct value 
Invalid OID	Error response 
Continuous polling	Stable 
Heartbeat update	Incrementing 

10. Performance Considerations

- Low memory usage
- Fixed MIB size
- Minimal CPU overhead
- Efficient UDP encoding

11. Limitations

- SNMPv1 only

- No SET support
 - No TRAP implementation
 - Limited OID capacity
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12. Future Improvements

- SNMPv2 / SNMPv3 support
 - Dynamic MIB
 - TRAP notifications
 - Improved lookup performance
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13. Industrial Relevance

This solution enables:

- Remote monitoring of devices
- Centralized system diagnostics
- Improved maintenance workflow

Applicable for:

- Airport logistics
 - Conveyor systems
 - Industrial automation
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14. Learning Outcomes

Technical:

- Embedded C programming
- UDP networking
- SNMP protocol
- BER encoding
- Multithreading

Conceptual:

- **System design**
 - **Protocol architecture**
 - **Debugging methodology**
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15. Conclusion

This project successfully implemented a functional SNMP agent for embedded systems.

It demonstrates:

- **Real-world communication protocol implementation**
- **Integration of networking with embedded systems**
- **Modular and scalable design**

The system provides a foundation for future expansion into advanced monitoring solutions.